HADOOP

Google File System

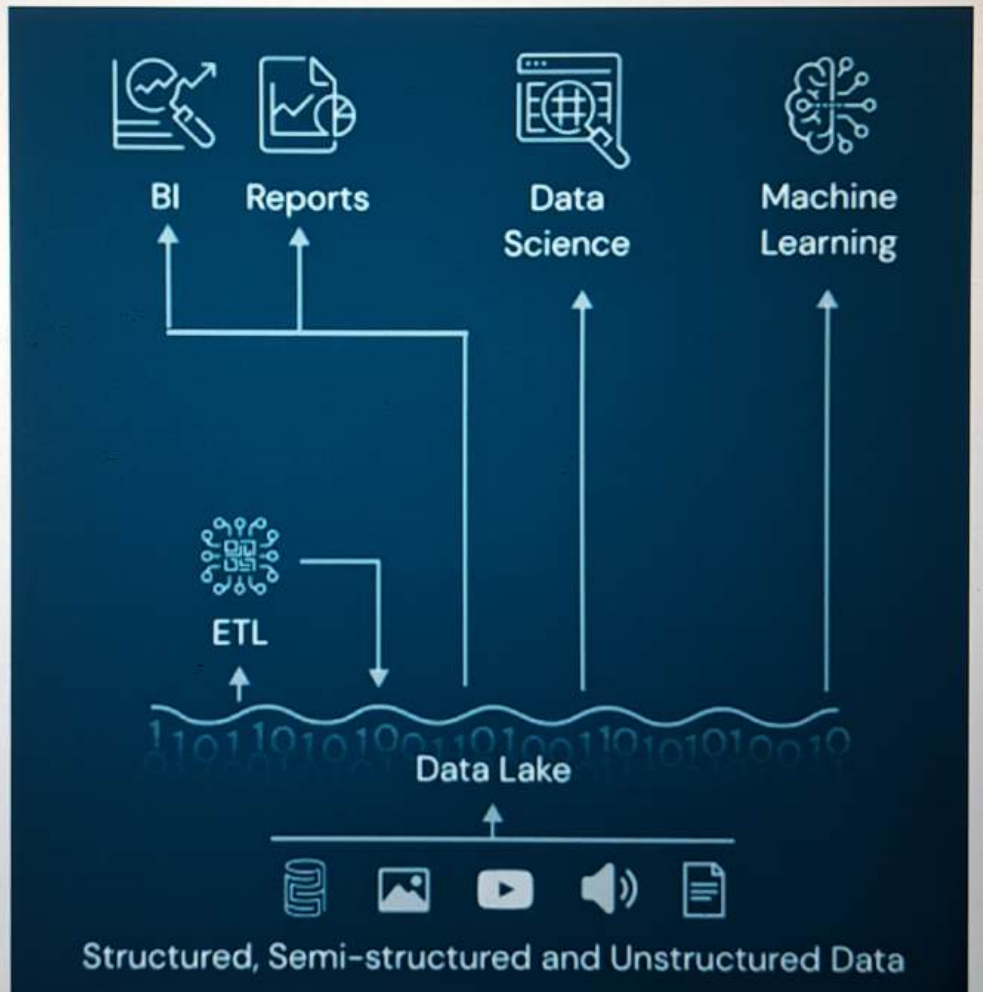
Hadoop Distributed File System

HADOOP

1. Teradata
2. Exadata

Ingest
Store/Process
Consume

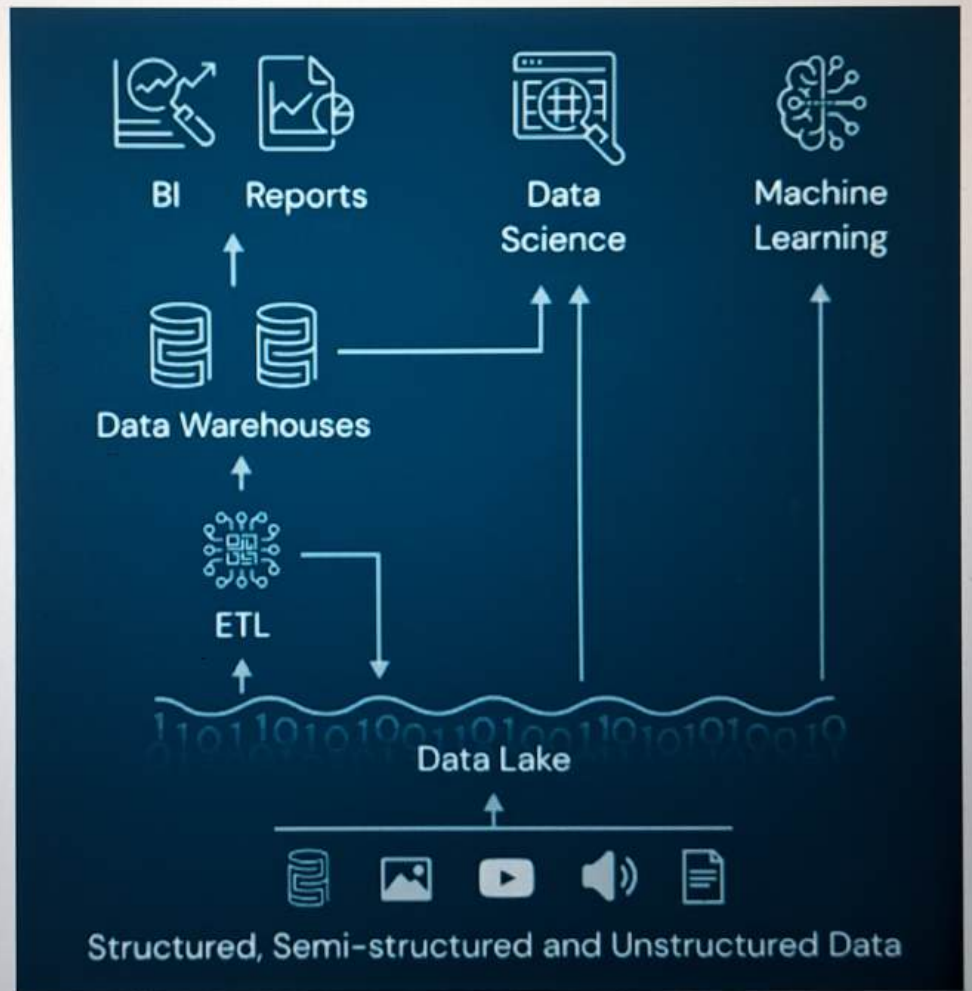
Data Lake



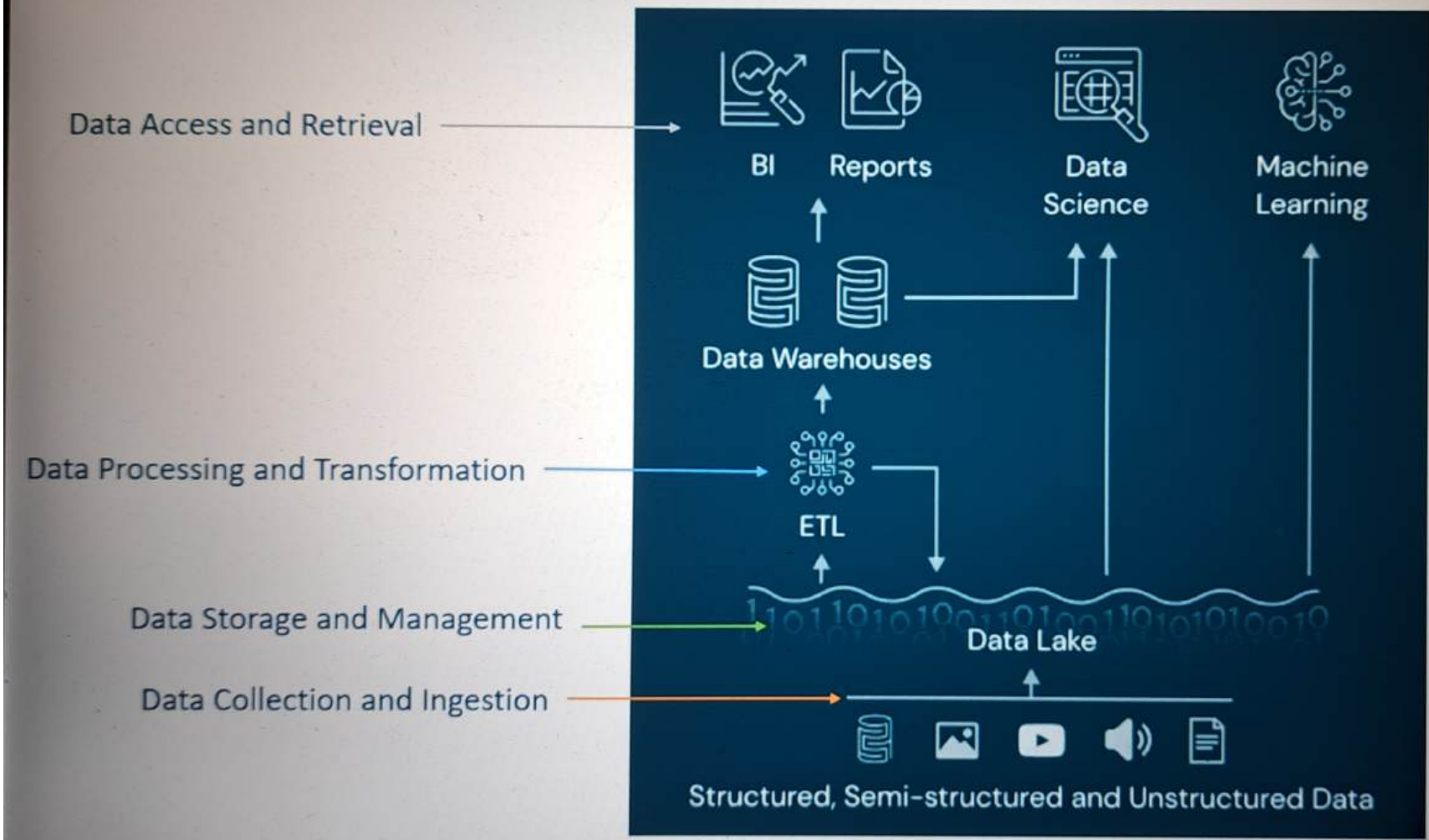
Key Missing Features

1. Transaction and Consistency
2. Reporting Performance

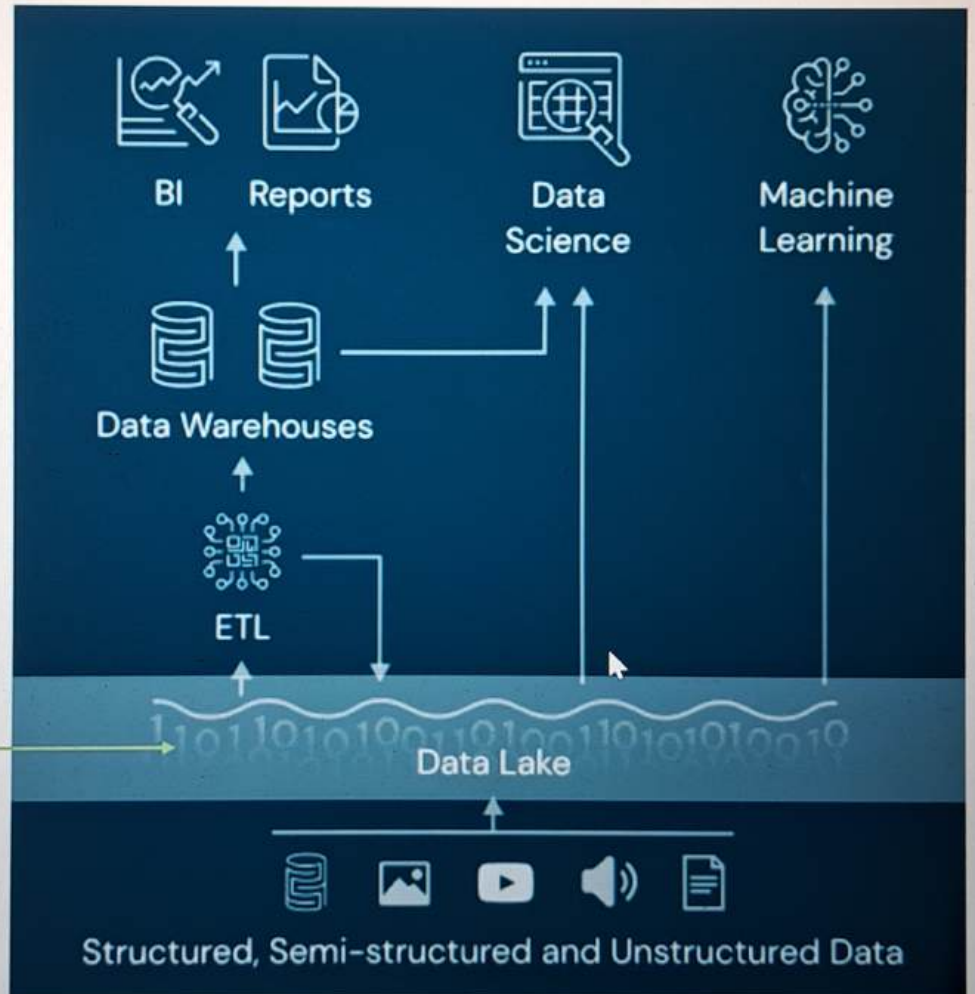
Data Lake


BI
Reports
Data Science
Machine Learning
Data Warehouses
ETL
Data Lake
Structured, Semi-structured and Unstructured Data

Data Lake

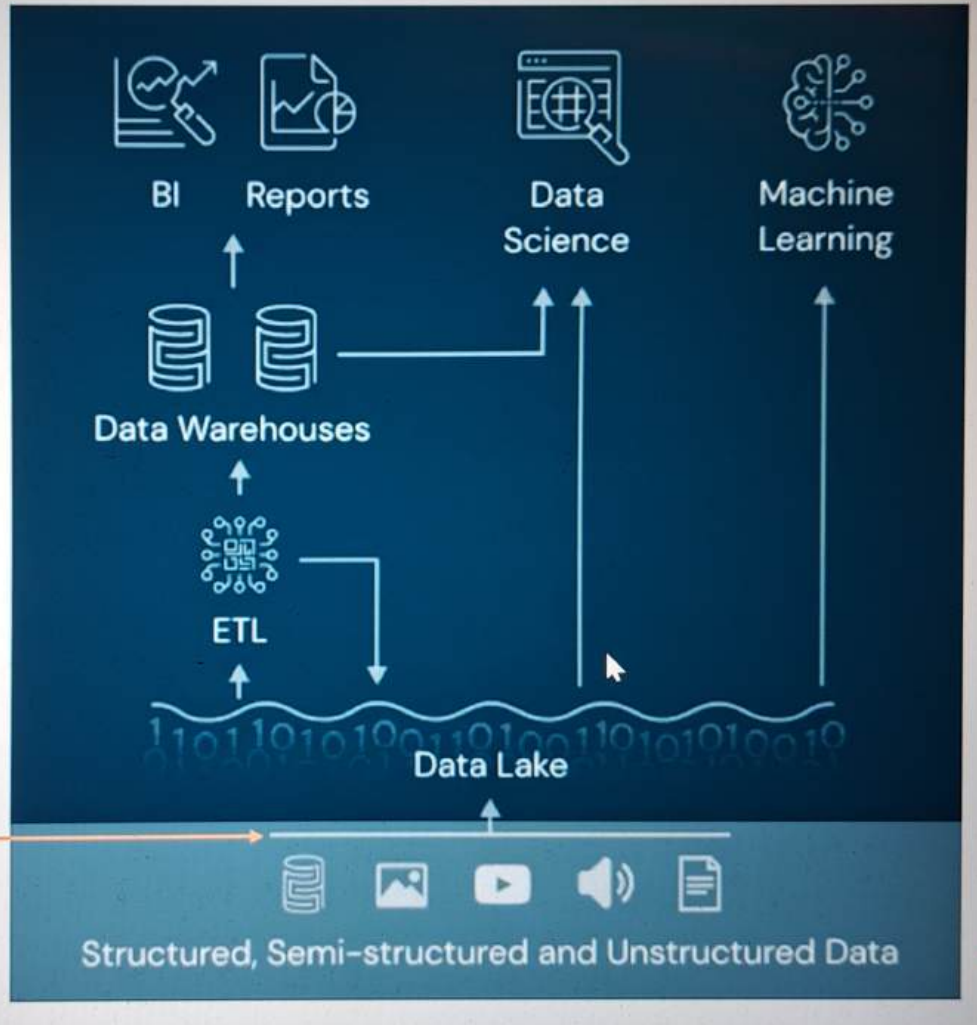

Data Access and Retrieval
BI
Reports
Data Science
Machine Learning
Data Warehouses
Data Processing and Transformation
ETL
Data Storage and Management
Data Lake
Data Collection and Ingestion
Structured, Semi-structured and Unstructured Data

Data Lake



Data Storage and Management

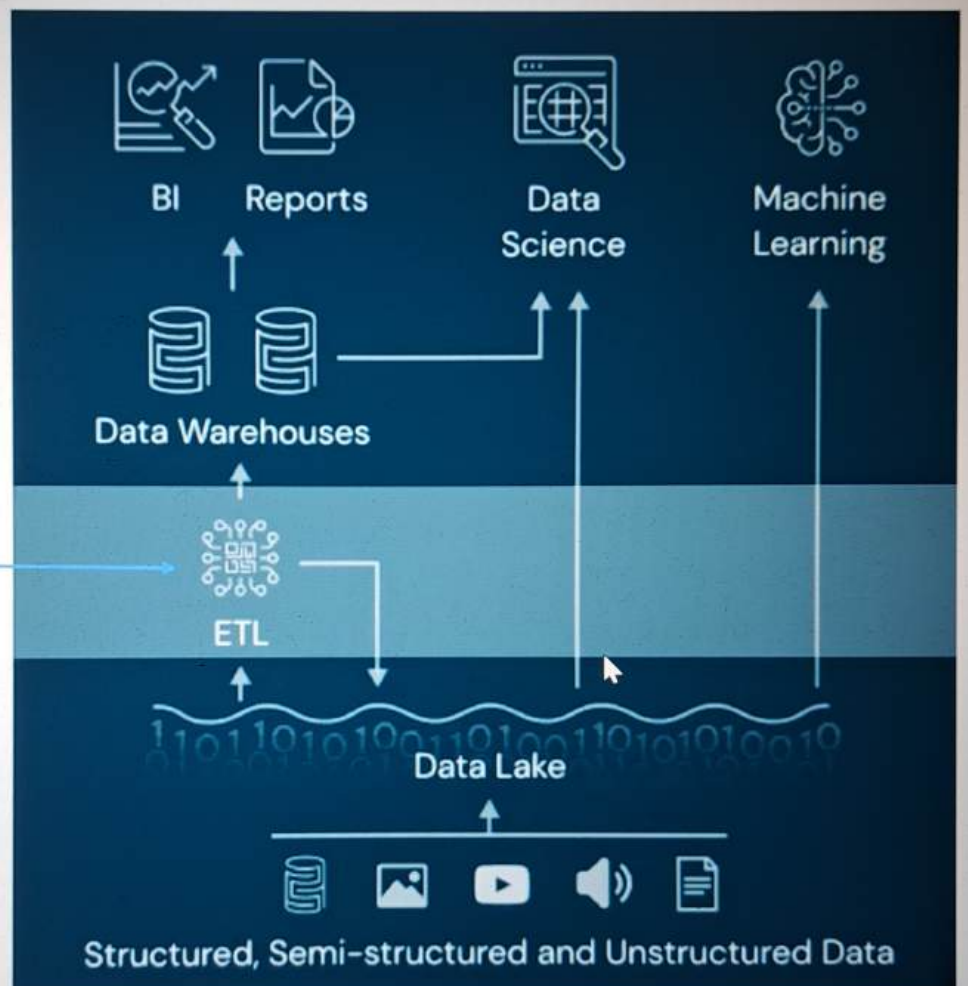
Data Lake



Data Collection and Ingestion

HVR
Azure Data Factory
Informatica
talend

Data Lake

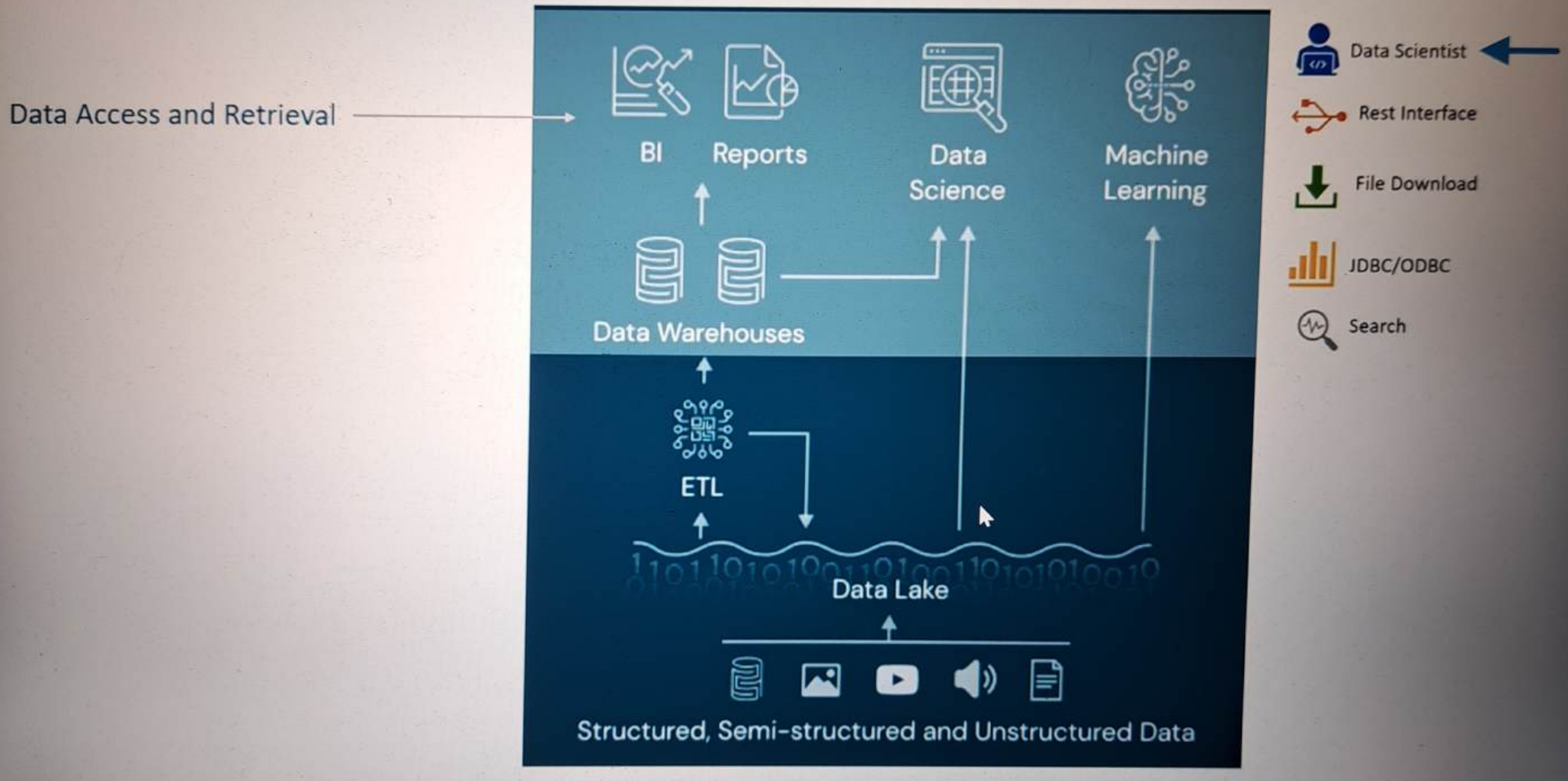


Data Processing and Transformation

1. Initial data quality check.
2. Transforming and preparing data.
3. Correlating, Aggregating, Analysing, and extracting business insights.
4. Applying machine learning model.

APACHE Spark

Data Lake


Data Access and Retrieval
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Machine Learning
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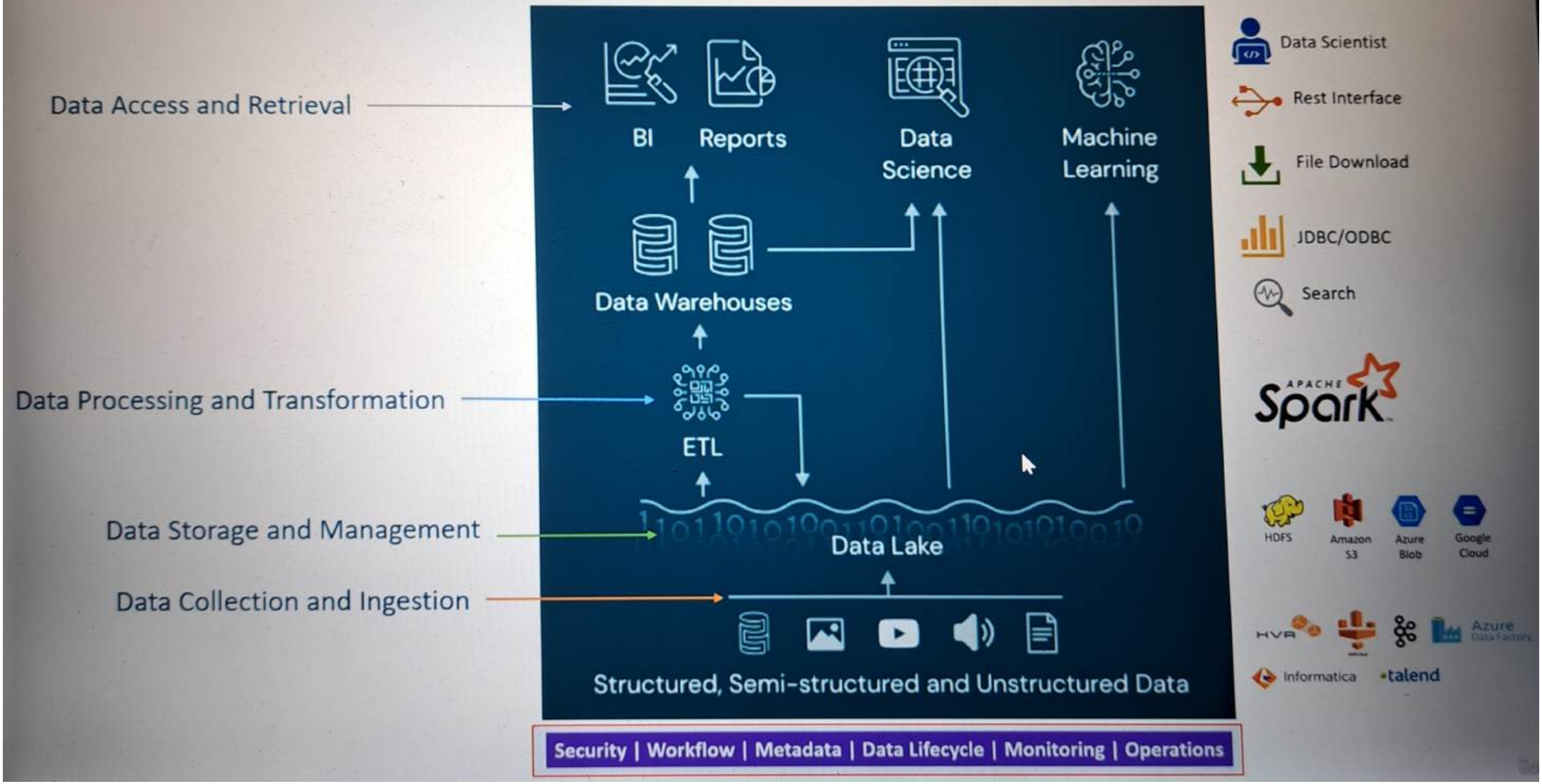
Data Lake
Structured, Semi-structured and Unstructured Data
Data Scientist
Rest Interface
File Download
JDBC/ODBC
Search

Data Lake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Structured, Semi-structured and Unstructured Data
Security | Workflow | Metadata | Data Lifecycle | Monitoring | Operations
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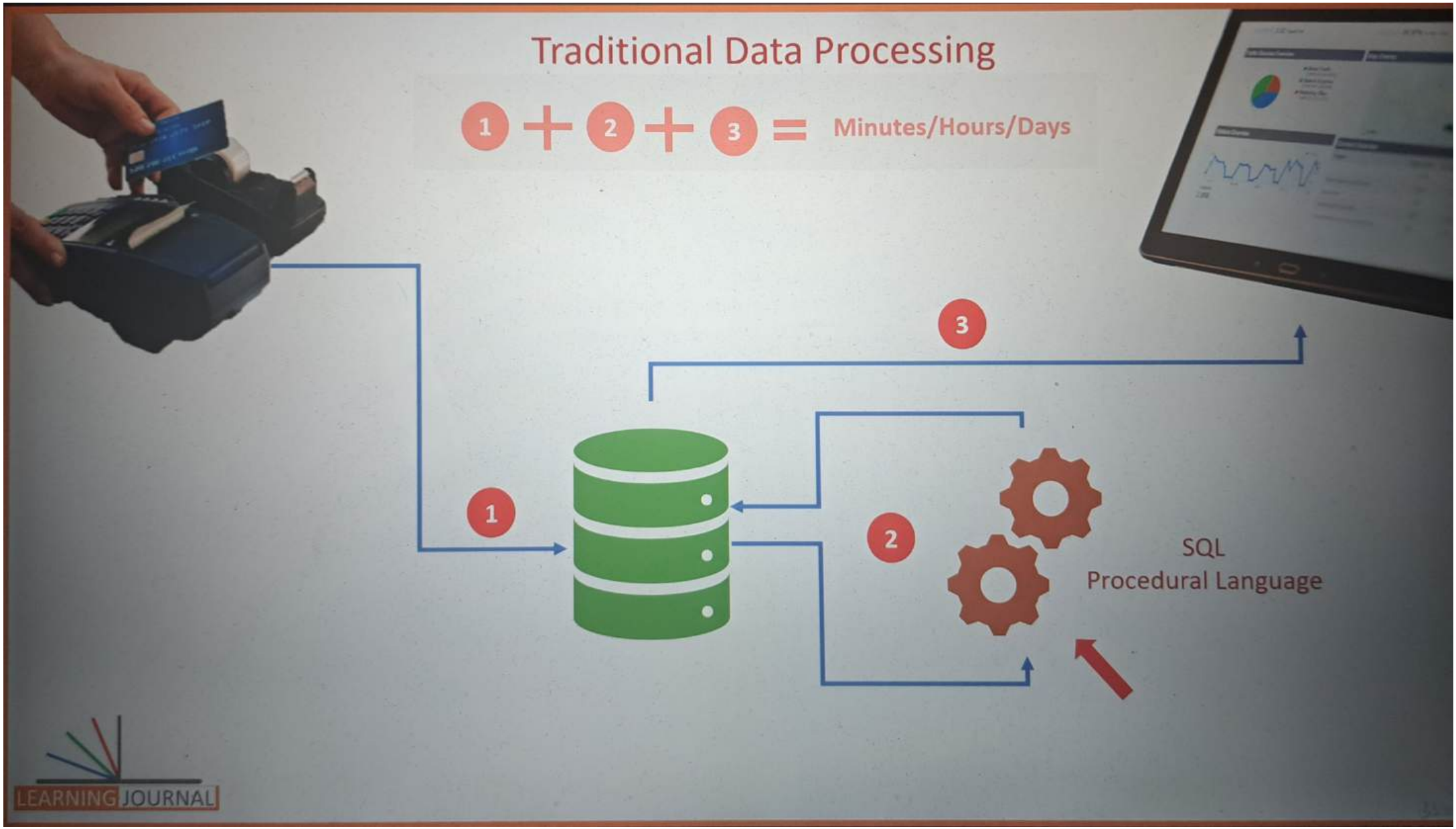
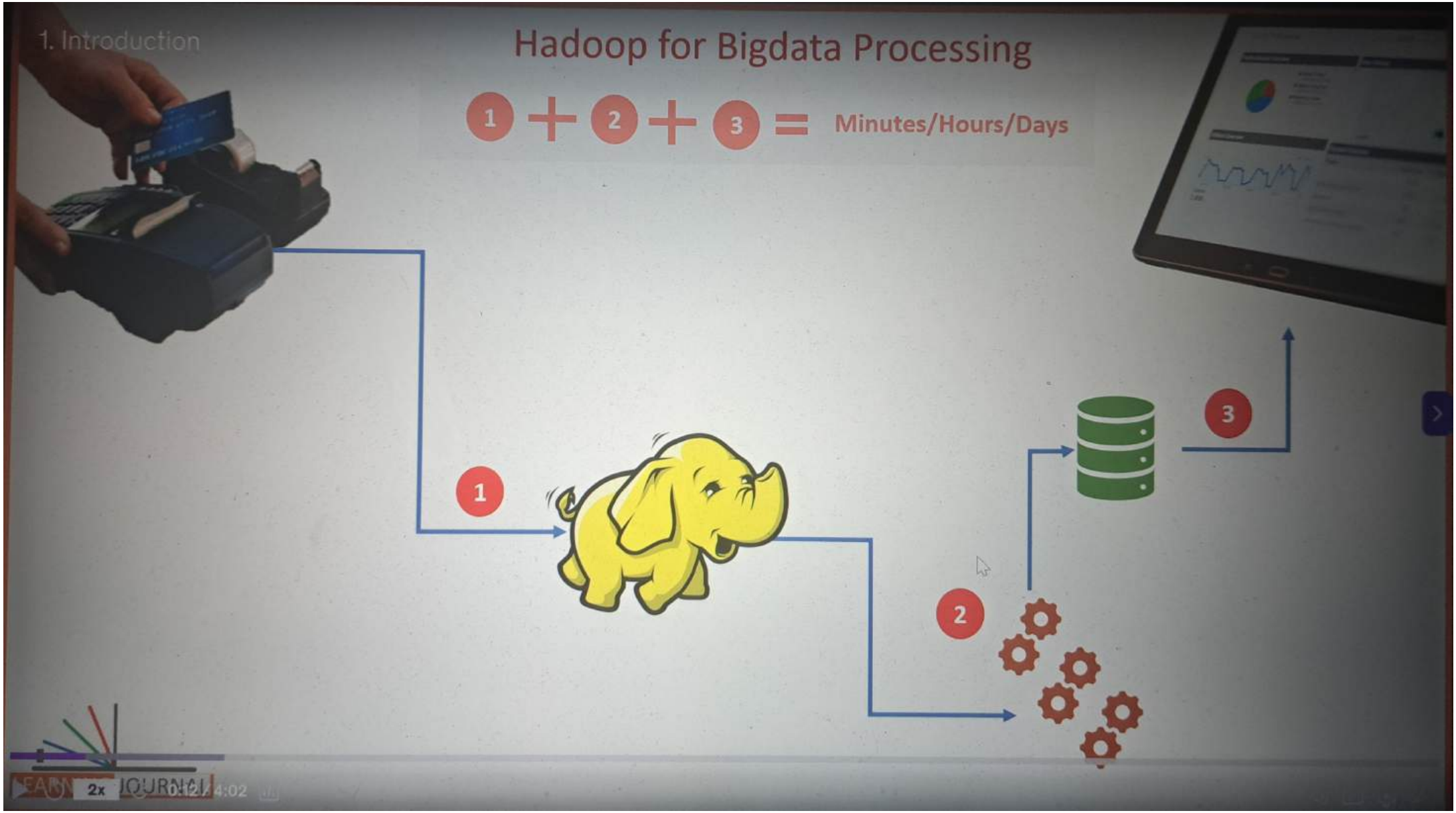
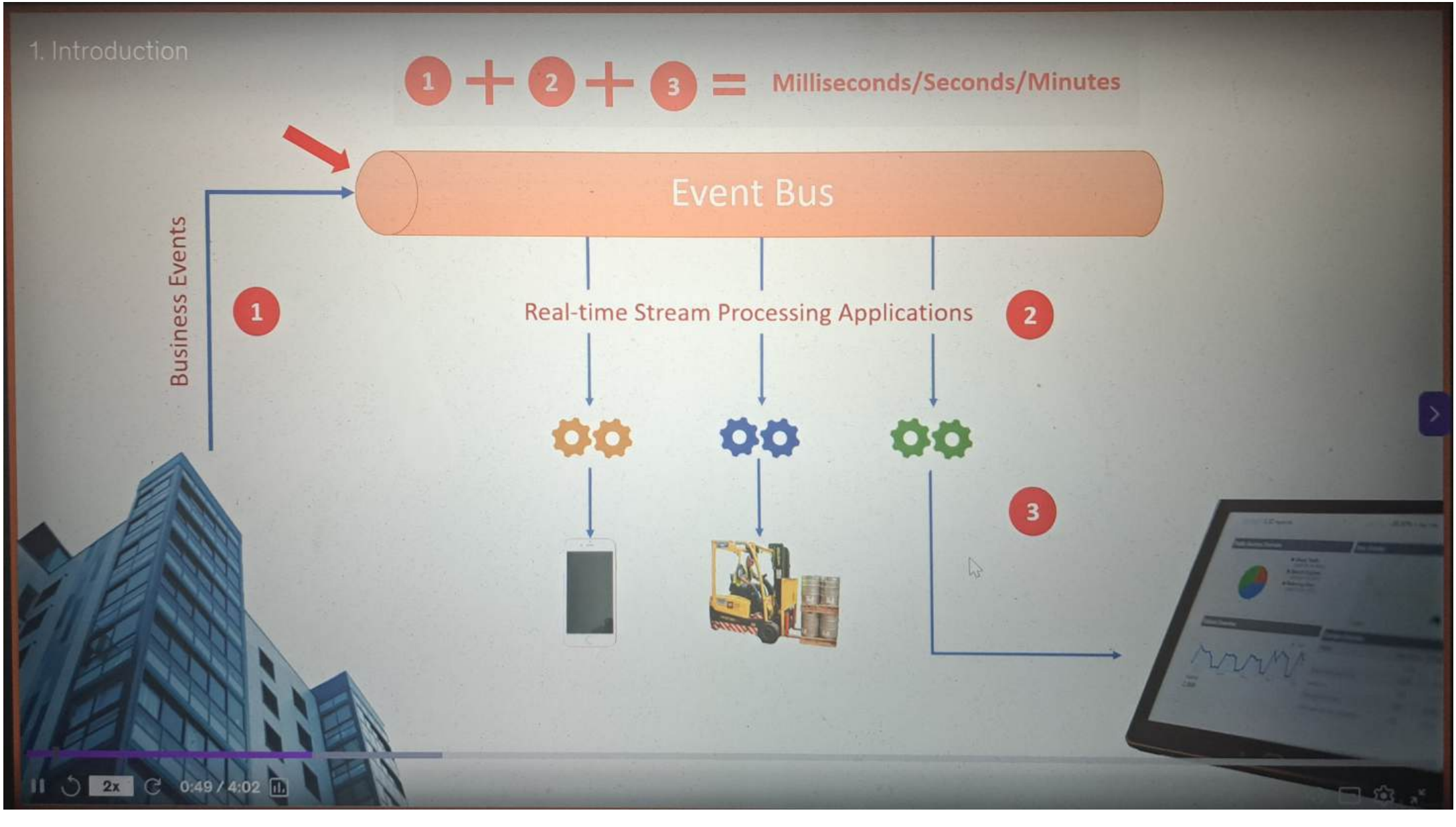
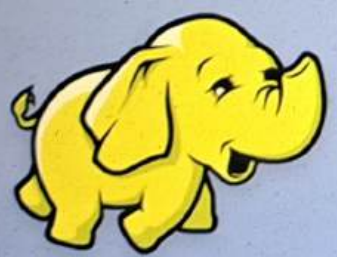
File Download
JDBC/ODBC
Search
APACHE Spark
HDFS
Amazon S3
Azure Blob
Google Cloud
HVR
Azure Data Factory
Informatica
talend


Traditional Data Processing
1 + 2 + 3 = Minutes/Hours/Days
1
2
3
SQL
Procedural Language
LEARNING JOURNAL


1. Introduction
Hadoop for Bigdata Processing
1 + 2 + 3 = Minutes/Hours/Days
1
2
3
2x


1. Introduction
1 + 2 + 3 = Milliseconds/Seconds/Minutes
Event Bus
Business Events
1
Real-time Stream Processing Applications
2
3
2x
0:49 / 4:02

1. Google File System
2. Map Reduce Model

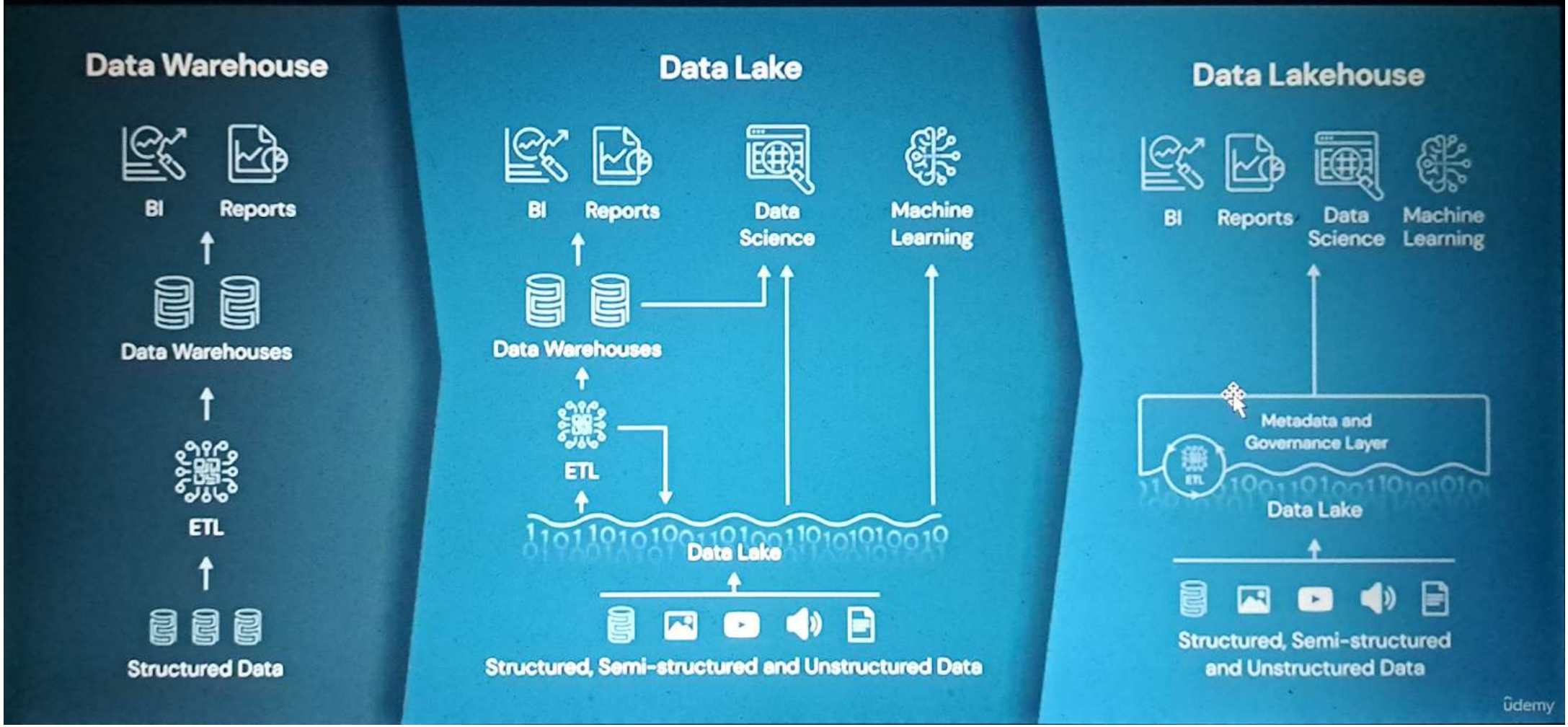


1. Hadoop Distributed File System
2. Map Reduce Model

Udemy

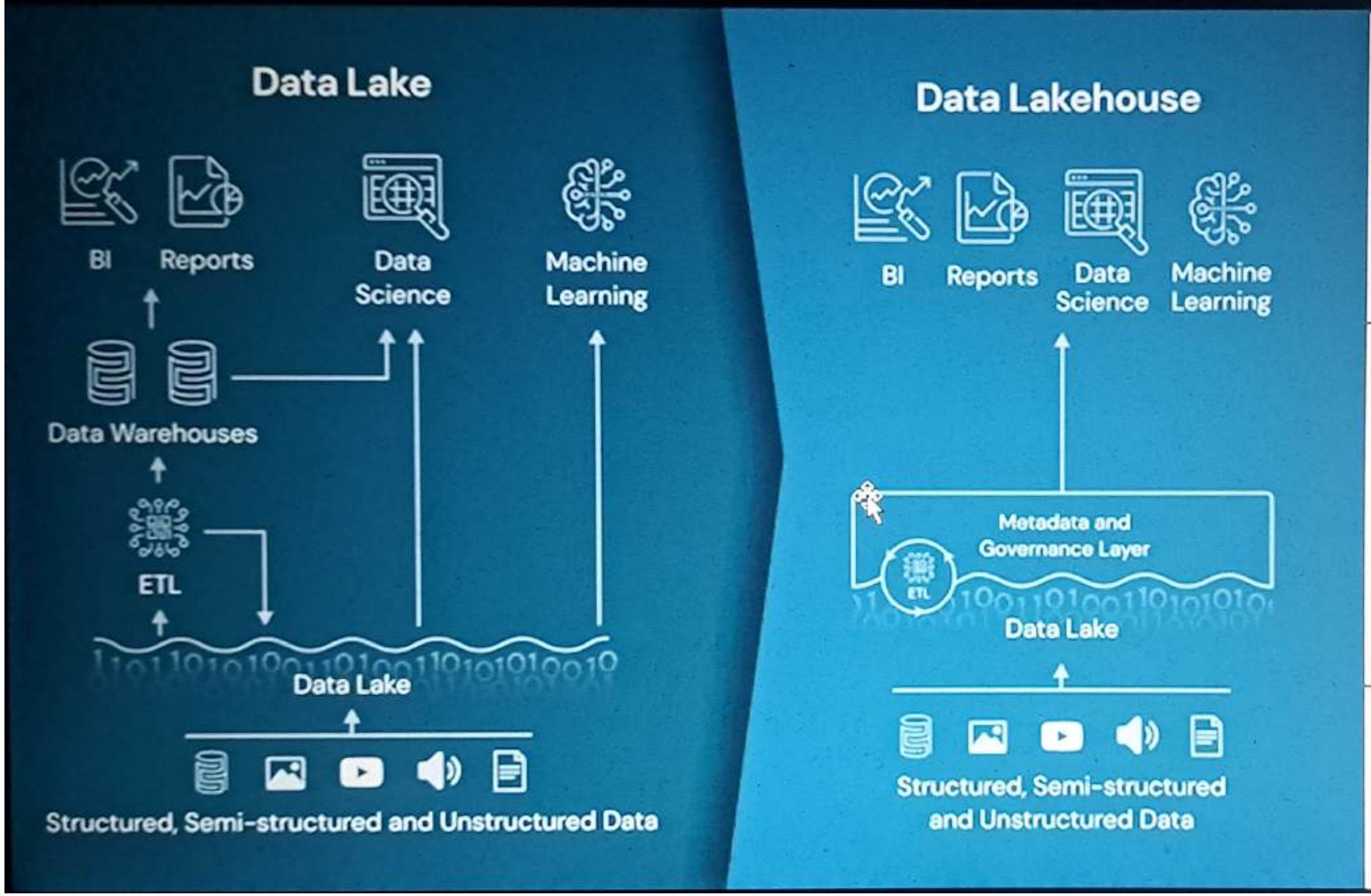
Data Engineering Architecture Evolution

ScholarNest



Data Engineering Architecture Evolution

ScholarNest



Various Tools and Technologies

databricks

APACHE Spark

Various Tools and Technologies